

WQD7025

DATA SCIENCE RESEARCH PROJECT

Session 2024/2025 Semester 2

Prediction of Extreme Weather Events: Machine Learning Models Trained on Multivariate Meteorological Data

Matric Number	23111363
Name	Wang Shibo
Occurrence	1
Supervisor Name	HANNYZZURA BINTI PAL AFFAL

Abstract

As global warming, the frequency and intensity of extreme weather events such as droughts, floods, and high temperatures have increased significantly, posing a serious threat to the safety of life and property across Asia. To enhance the ability to predict such events, this study proposes a hybrid CEEMDAN-SE-LSTM model that combines adaptive noise complete ensemble empirical mode decomposition (CEEMDAN), sequence reconstruction based on sample entropy (SE), and long short-term memory (LSTM) neural network for multivariate meteorological data-driven extreme weather forecasting.

The study selected monthly meteorological data from eight Asian countries from 1950 to 2023, including 15 important variables such as precipitation (pr), consecutive dry days (cdd), maximum daily precipitation (rx1day), and maximum temperature (tasmax). The modeling process follows the framework of "decomposition-reconstruction-prediction-integration": first, CEEMDAN decomposes each variable sequence into multiple intrinsic mode functions (IMFs) of different time scales; second, selects significant IMFs based on sample entropy values and reconstructs the sequence; third, these subsequences are input into the LSTM model for prediction; finally, the predicted subsequences are superimposed into a complete time series, and extreme weather labels (drought, flood, high temperature) are generated according to the preset threshold.

Compared with the traditional LSTM model, the performance evaluation of the CEEMDAN-SE-LSTM model shows that it has better ability in predicting complex and non-stationary meteorological series. This study not only overcomes the existing technical limitations in the field of climate prediction, but also provides a scientific basis for the implementation of the United

Nations Sustainable Development Goal 13 (climate action) and for the formulation of early warning systems and climate adaptation policies.

INTRODUCTION

1.1 Research Background

With the progress of global climate change such as global warming, long-term climate change has gradually led to the frequency of some extreme weather events becoming more and more frequent. The intensity of extreme weather events is also increasing, causing more serious consequences than before. According to the IPCC's 2023 Comprehensive Climate Report, human-induced warming has intensified extreme weather phenomena such as heat waves, heavy rains, droughts and storms around the world, causing widespread losses and damage [1]. Extreme heat events, in particular, caused massive death and disease in various locations due to severe heat waves in 2023 [1].

While the world is generally impacted by climate issues, Muhammad (2024) pointed out that countries in Asia may face even more severe challenges. Since Asia's average warming rate is higher than the global average, Asian countries are more affected by climate issues [2]. A report by the World Meteorological Organization (2022) pointed out that in 2021 alone, Asia experienced hundreds of major meteorological disasters, causing losses of approximately US\$36 billion and approximately 4,000 deaths, most of which were caused by floods [3]. The economic losses caused by floods in China that year were approximately US\$18.4 billion [3]. In Malaysia, floods also bring great harm. Malaysia experiences floods caused by heavy rains almost every year. For example, the Sabah floods in 2015 caused 241 deaths and losses of approximately US\$100 million. At the same time, Malaysia has also been affected by droughts due to climate instability, such as a drought in Kelantan in 2014 that caused an estimated US\$22 million in crop losses [4].

These rising patterns of loss of life and economic damage from floods, heat waves, and droughts highlight the urgency of research into predicting and preventing extreme weather.

The dramatic impacts of these climate issues have prompted more and more climate-related academic research, and various methods have begun to be applied to weather forecasting issues. Despite the extensive research on weather forecasting, most traditional extreme weather forecasting methods have inherent limitations in capturing the complex temporal and nonlinear characteristics of data [5]. For example, some meteorological agencies use statistical historical data, analyze historical data, and then use meteorology and physics to make predictions for weather and short-term climate forecasting, or use numerical forecasting to establish complex functional equations. These are essentially linear relationship processing methods. Simple statistical models, such as linear regression and static classifiers, usually assume that observations are independent of each other and have a linear relationship, and are therefore not suitable for highly dynamic climate systems [6]. Therefore, they have difficulty simulating long-term temporal dependencies or multivariate interactions, and may be overly sensitive to abnormal extreme values (outliers), resulting in poor generalization outside of training scenarios [6]. Overcoming the limitations of traditional technologies and updating models to enhance the ability to identify and predict extreme weather events, thereby improving the accuracy and reliability of extreme climate event predictions, is the current research trend.

1.2 Research Problem Statement

As the global climate continues to change, the demand for prediction and prevention of extreme weather events is increasing, especially in Asia, which is the most severely affected region. In recent years, heat waves, rainstorms, floods, and droughts have occurred frequently in Asia, causing great harm to

life and property in Asia. Such disasters are usually sudden and highly regionally diverse, so in order to make the prediction results more accurate, building a more intelligent and regionally adaptable prediction model has become a key objective [7].

Besides, some old traditional statistical methods, such as linear regression, have natural defects in dealing with nonlinear variable relationships and long-term time series dependencies, and static machine learning classifiers are also difficult to capture the dynamic evolution characteristics and variable interaction patterns in weather systems [8]. When faced with frequent outliers, unbalanced data, and complex cross-variable relationships, traditional methods usually show problems of overfitting or weak generalization ability. Despite growing datasets and improved forecasting methods, accurately predicting such events remains challenging due to the complexity and variability of climate systems. Therefore, it is essential to develop predictive models that integrate historical climate data and machine learning to better identify potential risks.

1.3 Research Questions

This study aims to provide insight into the following research questions:

1. How to utilize historical climate data to predict the occurrence of droughts, floods, and heatwaves in Asian countries?
2. For each type of extreme weather problem, what are the most influential meteorological characteristics (e.g., cdd, pr, rx1day, tasmax, hours)?
3. What are the advantages of the CEEMDAN-SE-LSTM model over the traditional LSTM model in predicting weather data with multi-time scale characteristics such as seasonality and periodicity?

1.4 Research Objectives

1. To design and evaluate the predictive performance of a hybrid CEEMDAN-SE-LSTM model for forecasting three major types of extreme weather events—droughts, floods, and heatwaves—using monthly climate indicators from eight Asian countries spanning 1950 to 2023.
2. To identify and assess the most influential meteorological variables contributing to the prediction of extreme weather events through correlation analysis, feature selection techniques, and cross-country variable importance evaluation.
3. To validate the robustness and generalizability of the CEEMDAN-SE-LSTM model by benchmarking its forecasting accuracy and stability against baseline models, including LSTM, EEMD-LSTM, and CEEMDAN-LSTM.

1.5 Research Scope and Contribution

This study aims to develop a CEEMDAN-SE-LSTM hybrid predictive model based on multivariate meteorological data to forecast extreme weather events such as droughts, floods, and heat waves in Asian countries. The scope of the research is defined as follows:

1. Time Scope: Monthly meteorological observation data spanning from 1950 to 2023 are used for model training and testing, with the prediction target focusing on extreme weather trends starting from 2024 and beyond.
2. Geographical Scope: Eight climate-sensitive countries in Asia—Malaysia, China, India, Indonesia, Japan, Thailand, the Philippines, and Vietnam—are selected as the focal regions of this study.
3. Variable Scope: A total of 15 key meteorological variables are included, such as precipitation (pr), consecutive dry days (cdd), extreme maximum temperature (tasmax), and maximum daily rainfall (rx1day), among

others, to support the prediction of three major types of extreme weather events.

4. Data Source: All meteorological data are obtained from the World Bank Climate Change Knowledge Portal.
5. Technical Scope: The CEEMDAN method is primarily employed for time series decomposition; sample entropy (SE) is integrated to identify informative subsequences; and the LSTM neural network is applied for modeling and prediction.

The main academic and practical contributions of this study are summarized as follows:

1. Methodological Innovation: A CEEMDAN-SE-LSTM model is developed based on the "decomposition-reconstruction-prediction-integration" framework. This approach systematically enhances the generalization capability and predictive accuracy of traditional LSTM models in meteorological forecasting, particularly for climate time series characterized by non-stationarity and multi-scale features.
2. Filling Research Gaps: In contrast to existing studies that typically focus on a single type of extreme event, a single geographic region, or conventional modeling approaches, this research simultaneously addresses three types of extreme weather events and integrates the LSTM model with CEEMDAN-SE decomposition-reconstruction techniques, offering novel technical insights for climate forecasting methodologies.
3. Support for Climate Action (SDG 13): This study aligns with the United Nations Sustainable Development Goal 13 (Climate Action) and contributes to climate resilience by enhancing the capacity to predict extreme weather events. The potential impacts include:

At the societal level: Facilitating early warning systems for governments and communities, thereby reducing risks to life and property.

At the economic level: Providing decision-making support for sectors such as agriculture and water management, helping to mitigate economic losses caused by severe weather.

At the national policy level: Offering scientific evidence for climate-related policies, adaptive regulations, and the development of national meteorological monitoring systems.

2. LITERATURE REVIEW

2.1 Overview of Detection Methods

At present, the time series prediction methods mainly used to predict precipitation and other meteorological data include traditional moving average (MA), autoregressive (AR), autoregressive moving average (ARMA) and artificial neural network (ANN) based on machine learning. Compared with the above methods, the LSTM can handle non-stationary sequences, is widely used in weather forecasting, and has high accuracy in short-term climate data forecasting.

2.2 Comparative Analysis of Methodologies

A wide spectrum of methodologies has been employed in recent years to forecast extreme weather events. These methods range from traditional statistical approaches to more advanced machine learning and deep learning frameworks. This section provides a structured comparative analysis of 10 key studies relevant to droughts, floods, and heatwaves prediction, especially in multivariate, time-series, and meteorological contexts. The analysis highlights the strengths, limitations, and distinctive contributions of each method in context-specific climate forecasting scenarios.

Islam et al. focused on the annual yield prediction of Aman rice, the staple food of Bangladesh, to support the formulation of future agricultural policies in

Bangladesh and promote the goal of SDG 12. Using 52 years of historical climate data from 1970 to 2022, including temperature, rainfall, humidity, wind speed, etc., they compared the accuracy of crop yield prediction using random forest, neural network, decision tree and linear regression. The comparison found that random forest performed best, with an R^2 of 0.964 and an RMSE of 5.44, making it the most suitable model for predicting rice yield under the influence of climate change. The random forest model can effectively deal with high-dimensional meteorological data and nonlinear relationships between variables, and has strong robustness. However, since it does not use a time series model, the model cannot capture long-term trends between years [9].

Kumar et al. focused on predicting urban rainfall in Delhi, India, to support flood warning and related urban water resource management policies. The experimental data was daily rainfall in Delhi, India from 1980 to 2021, and compared the rainfall prediction performance of various machine learning models, including CatBoost, XGBoost, LGBM, Lasso, Ridge, and Linear Regression. The experimental results showed that the XGBoost model performed better than other models, with $MAE \approx 0.02$, $RMSE \approx 0.10$, and $R^2 \approx 0.99$. XGBoost showed excellent performance in modeling nonlinear and time series features, but its research was only based on the Delhi region and lacked cross-regional generalization modeling [10].

Wang et al.'s study focused on drought prediction in the Dongjiang River Basin of China. They used a single-layer LSTM structure and used 10 types of meteorological and environmental data to predict future meteorological variables and calculated the SPEI (Standardized Precipitation Evapotranspiration Index). The model combines multi-source meteorological and geographical factors, but the model is limited to the Dongjiang River Basin of China, and its generalization ability remains to be verified. In

addition, LSTM was not compared with other machine learning models in the experiment [11].

Zhang et al.'s project involved predicting the severity of drought at 32 weather stations in Shaanxi Province, China, and predicting the SPEI values for the next month. In addition, they also set up classification to determine whether it was moderate, severe, or extreme drought, and compared the performance of DLNM, ANN, and XGBoost models. XGBoost performed best, with an SPEI prediction accuracy of 89%-97% and a classification prediction accuracy of 76%–94%, while ANN and DLNM performed poorly in predicting extreme drought, and the ANN model had poor stability and required multiple initializations to average the results [12].

Li et al. combined the simulated data output by the NAM model with the measured data of real extreme flood events to construct a mixed training set for 1-6 hour flood lead time prediction of the LSTM model. Their project mainly compared the difference in the impact of using pure measured data, simulated data, and real simulated mixed data on the prediction performance of the training LSTM model. The LSTM model with the best final effect had a MAE of 59.97-91.59. Their project verified the impact of the structure of the test data on the prediction effect, but did not compare the effects of different models from the perspective of model selection. It only demonstrated the feasibility of the LSTM model in predicting flood extreme events on time series data [13].

Shafiq et al. compared three deep learning models, ANN, CNN, and LSTM, to predict extreme high temperatures in the Pakistani city of Lahore. The LSTM model performed best, with an accuracy of 96.2%, better than ANN (89.1%) and CNN (92.5%). Before modeling, they used XAI analysis to find that maximum temperature and humidity are the most important factors in predicting heat waves, which improves the interpretability of the model. However, the data set is small, and the regional scope is only the city of

Lahore, and the model's ability to generalize across regions has not yet been verified [14].

Essa et al. predicted thunderstorms in South Africa. In this project, they compared three LSTM hybrid models, LSTM-FC, CNN-LSTM, and ConvLSTM. All models used the historical meteorological and lightning data of the past 240 hours as input to predict the frequency of lightning strikes in the next hour, thereby quantifying the intensity of thunderstorms. By comparing the experimental results, the CNN-LSTM model performed best in prediction results, which can extract both temporal and spatial features at the same time and is the most suitable architecture for the current regional model. However, the data did not introduce noise influence, and overfitting problems may occur [15].

Dikshit et al. compared the results of LSTM and ARIMA in the SPEI drought prediction in New South Wales, Australia. LSTM performed better than ARIMA, especially in capturing extreme droughts. ARIMA is more suitable for short-term trends, while LSTM can more effectively restore long-term trends. In addition, their project was trained using a century of climate data (1901–2018), and the model has strong generalization. Their project highlights the advantages of deep learning such as LSTM in modeling nonlinear and long-memory climate data. However, there are still shortcomings in their project, such as not using explanatory analysis tools to explain the importance of variables [16].

Xie et al. proposed an innovative model that divides the water level changes during a flood into multiple time periods and constructs vector directions of rising, falling, and stable. By combining vector direction with LSTM, a VD-LSTM model for flood forecasting was proposed. Their research aims to improve the responsiveness of flood prediction models to flood and recede processes. The model is trained using actual flood event data from the Jingle and Lushi basins of the Yellow River in China from 1980 to 2016. The model

introduces "vector direction" information for the first time, which enhances the model's sensitivity to the time series of flood and recede processes, but there are still deviations in the simulation of the peak section of a bimodal flood [17].

Frifra et al. combined LSTM and XGBoost for storm prediction. LSTM is used to predict multiple continuous feature variables of storms, while XGBoost is used for binary classification problems, outputting a label (0 or 1) of whether a storm has occurred, which is used to predict the date of the storm. The study area is the western coastal area of France, and the data comes from the measured meteorological data of the Brittany buoy station from 1996 to 2020. The experimental results show that LSTM has obvious "extreme value missed detection" in storm feature prediction. For example, the values of feature values such as wind speed and wave height are underestimated when the storm occurs. The traditional LSTM model is more suitable for the prediction of relatively stable meteorological factors, such as temperature, air pressure and other features, with higher prediction accuracy [18].

Study	Contributions	Model	Data	Strength	Weakness
Islam et al. (2024) [9]	Predict Aman rice yield using 52 years of climate	Random Forest	Temp, rainfall, humidity, wind (1970–2022)	High accuracy ($R^2=0.964$), handles complex data well	Not time series; misses long-term trends
Kumar et al. (2023) [10]	Predict daily rainfall in Delhi for flood warning	XGBoost	Daily rainfall (1980–2021)	Very accurate ($R^2\approx0.99$), good at time series features	Only tested in Delhi; lacks generalization

Wang et al. (2023) [11]	Drought prediction in Dongjiang River	LSTM	10 weather + environment variables	Uses multi-source data	No model comparison; only tested in one region
Zhang et al. (2019) [12]	Drought level classification using SPEI in Shaanxi	XGBoost	SPEI at 32 weather stations	High prediction and classification accuracy	Other models (ANN, DLNM) unstable or weak
Li et al. (2024) [13]	Flood forecast using mixed real + simulated	LSTM	NAM + real flood events, 1–6 hr lead	Mixed data improves model	No comparison with other models
Shafiq et al. (2025) [14]	Heatwave prediction in Lahore using deep learning	LSTM (compared with ANN, CNN)	Max temp, humidity + XAI	Best accuracy (96.2%); interpretable	Small dataset; only for one city
Essa et al. (2022) [15]	Thunderstorm prediction in South Africa	CNN-LSTM	240-hour lightning + weather data	Best at capturing time + space info	No noise data; possible overfitting
Dikshit et al. (2021) [16]	SPEI drought prediction in Australia	LSTM vs ARIMA	100+ years of climate data (1901–2018)	Good at long-term, extreme drought prediction	Lacks variable explanation tools
Xie et al. (2024) [17]	Flood forecasting using vector directions	VD-LSTM	Flood stage vectors (rise/fall/stable) with other environment variables	Sensitive to flood process changes	Peak flood simulation not accurate

Frifra et al. (2024) [18]	Storm prediction in coastal France	LSTM + XGBoost	Wind, wave data from buoy station	Combines prediction + classification	LSTM underestimates extreme values
---------------------------	------------------------------------	----------------	-----------------------------------	--------------------------------------	------------------------------------

These literatures show the diversity of current methods used for extreme weather prediction. The models used include machine learning models such as XGBoost and RF, as well as deep learning architectures such as LSTM, and some hybrid LSTM models such as CNN-LSTM and VD-LSTM. The experimental results show that different models show obvious differences in adaptability when dealing with different types of extreme climate events. For example, XGBoost shows excellent precision and recall in storm classification, while LSTM is more suitable for capturing long-term climate trends dominated by time series such as droughts and heat waves. Among them, various hybrid models that optimize LSTM on the corresponding problems perform very well, and the prediction accuracy and recall are generally better than traditional LSTM models. Therefore, for the long-term climate characteristic trends dominated by time series such as precipitation, temperature, and humidity involved in this project, the project will optimize on the basis of LSTM and build a CEEMDAN-SE-LSTM model.

2.3 The application of CEEMDAN-SE in time series problems

Because meteorological data such as precipitation and temperature are influenced by various factors including hydrology, atmospheric circulation, monsoons, terrain, and human activities, meteorological data time series tend to exhibit strong non-stationary random fluctuations as well as complex features such as modal aliasing and multiple time scales.

For the reasons mentioned above, the prediction error of traditional time series methods is usually relatively large. Huang et al. (1998) proposed a new method for dealing with nonlinear and non-stationary time series - the Empirical Mode Decomposition (EMD) method. This method can adaptively decompose the time frequency based on the local time-varying characteristics of the data itself and perform stationary processing on the sequence [19].

However, EMD is inevitably prone to the phenomenon of Intrinsic Mode Function (IMF) aliasing. Modal aliasing not only makes it impossible to accurately represent the IMF components at each time scale, but may even cause individual IMF components to lose their physical meaning. In order to eliminate the modal aliasing problem as much as possible and keep each IMF component generated by decomposition physically unique, Wu and Huang (2009) proposed the Ensemble Empirical Mode Decomposition (EEMD) filtering method. This method has suppressed the modal aliasing problem to a certain extent and achieved good results in climate prediction. However, there is residual noise in the IMF component after EEMD decomposition, and each time, different amplitudes of adding white noise need to be subjectively selected [20].

Therefore, Torres et al. (2011) proposed a fully adaptive method called Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN). This method effectively overcomes the nonzero reconstruction error problem of EEMD by adding adaptive white noise at each stage, ensuring the completeness of the decomposition. It also reduces the number of iterations and improves computational efficiency [21].

Liu et al. (2023) proposed a hybrid CEEMDAN-SE and Bidirectional LSTM Neural Network model when predicting wind power data, because wind power data has strong randomness, non-stationarity and high complexity. The wind power data of a wind farm in Hami, Xinjiang in 2021 was decomposed into

multiple IMF sequences and then reconstructed into 5 subsequences.

BiLSTM was used to predict the 5 subsequences separately. This design can effectively solve the multimodal and non-stationary problems of the original wind power data, solve the problem of mutual interference of different time scale features when predicting wind power data, and improve the accuracy of wind power prediction.

2.3 Literature gap

From the above literature, we can find that although there are many machine learning and deep learning methods such as LSTM, XGBoost, CNN-LSTM, etc. in the field of extreme weather prediction, there are still several research gaps. First, many of the above literatures only focus on a certain type of extreme weather, such as thunderstorms or floods. There is still a lack of models with a unified architecture to output multiple extreme weather labels. Secondly, most studies are only in a single city or region. For example, the study by Zhang et al. only predicts the drought level in Shaanxi city. There is still a lack of universality of analytical models in different countries and meteorological types. Third, meteorological time series have strong non-stationary and multi-scale fluctuation characteristics. Existing studies such as the study by Li and Shafiq et al. only use LSTM to capture the long-term change trend of climate data. Although the results have high accuracy, the model does not capture the short-term disturbances of climate data and the fluctuation characteristics at multiple time scales, resulting in insufficient prediction capabilities in scenarios where extreme event turning points or extreme weather problems occur.

This project proposes a combined prediction model CEEMDAN-SE-LSTM based on the idea of "decomposition-reconstruction ensemble", leveraging the advantages of CEEMDAN's better decomposition , LSTM's higher long-term

prediction accuracy, and SE's faster reconstruction efficiency. CEEMDAN will be used to decompose the original sequence, obtain IMFs of different time spans (quarterly, annual, and multi-year), and extract dynamic features at each frequency level. For example, quarterly IMF can reveal short-term climate change, annual IMF can represent seasonal changes, and multi-year IMF can show long-term trends. This can eliminate non-stationary interference and enable subsequent LSTM models to more accurately identify the precursors and trends of extreme events. Subsequently, the forecast results of each IMF are reconstructed into an overall output through stacking integration (SE), thereby retaining sudden abnormal signals and improving model accuracy.

3. RESEARCH METHODOLOGY

3.1 Research Workflow

In order to enhance the prediction accuracy and interpretability of non-stationary meteorological time series, this study proposes a hybrid model based on CEEMDAN, sample entropy (SE), and LSTM. Initially, key meteorological features are selected through feature screening. Subsequently, the CEEMDAN method is applied for multi-scale decomposition, followed by reconstruction and dimensionality reduction using sample entropy (SE). Finally, the LSTM model is employed to train each reconstructed component individually, thereby constructing an extreme weather forecasting model. The overall methodology is illustrated in Figure 3.1.

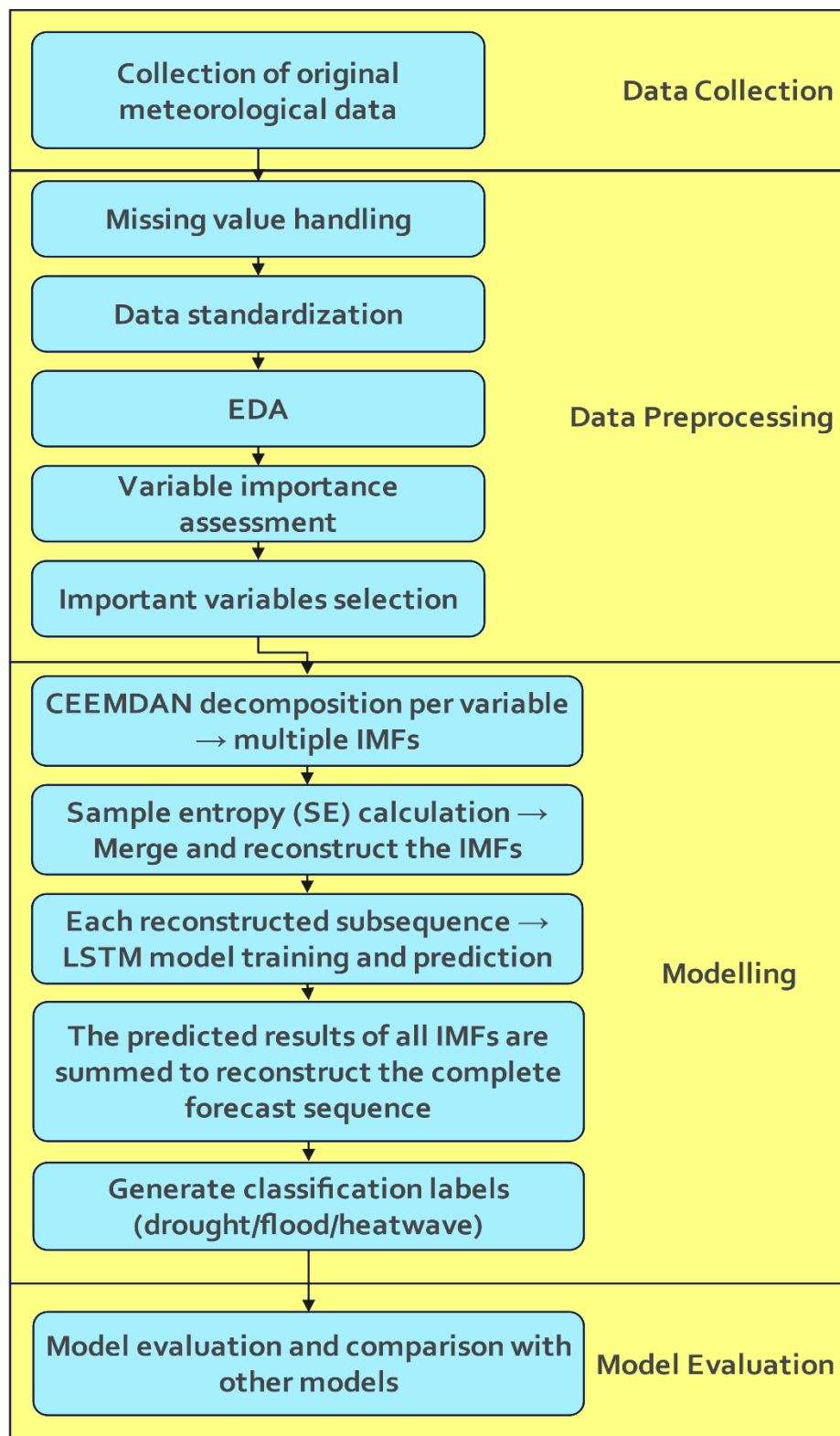


Figure 3.1.

3.2 Data Collection

Data were gathered from publicly available climate archives and portals, Climate Change Knowledge Portal (World Bank). From this source I extracted monthly values for all relevant variables between 1950 and 2023 for the eight target countries (Malaysia, China, India, Indonesia, Japan, Thailand, Philippines, Vietnam). The selected variables align with the study's extreme-event focus: total precipitation (pr), consecutive dry days (cdd), temperature extremes (tasmax, tasmin), relative humidity (hurs), monthly rainfall aggregates (pr), maximum 1-day and 5-day rain (rx1day, rx5day), and counts of heavy rain days (r20mm, r50mm), days above high-temperature thresholds (hd35, hd40), heat indices (hi35, hi37, hi39), and daily temperature metrics (tasmax, tasmin). Download climate data in separate CSV format for each variable, with each file corresponding to monthly records of one variable for 8 specific countries. Merge these datasets using country and timestamp (year-month) as key identifiers, resulting in a single time series dataset containing all variables, 8 countries $\times$ 74 years $\times$ 12 months $\times$ 15 variables.

3.3 Data preprocessing

In order to enhance the adaptability and predictive accuracy of the model for extreme weather events, this study systematically preprocessed multivariate meteorological data spanning from 1950 to 2023 prior to model development.

The preprocessing workflow consisted of five key steps: missing value handling, data standardization, exploratory data analysis (EDA), variable importance evaluation, and selection of key variables.

(I) Missing Value Handling

Given that the dataset encompasses meteorological records from multiple countries over a long time period, certain variables and time points may

exhibit missing values. To address this issue while preserving temporal and seasonal characteristics, different imputation strategies were applied. For short-term data missing (less than three consecutive months), linear interpolation was employed. For longer continuous segments of missing data, values were imputed using the multi-year historical average for the corresponding month and country, thereby minimizing potential data distortion.

(II) Data Standardization

Due to the inclusion of 15 distinct meteorological variables with varying units and scales, standardization was essential to ensure comparability and prevent bias toward variables with larger magnitudes during model training. This study applied minimum-maximum normalization (Min-Max Scaling) at the country level, transforming all variables into the range $[0,1]$. This approach ensures uniformity across variables and enhances the reliability of subsequent analytical and modeling processes.

.(III) Exploratory Data Analysis (EDA)

Following standardization, each variable undergoes a series of analyses including distribution testing, trend analysis, and extreme value identification. Time series trend plots and box plots are employed to examine the temporal variation characteristics of the variables. This process enables the preliminary identification of periodic patterns, anomalies, and seasonal fluctuations within the data, thereby establishing a foundation for subsequent feature selection and model development.

(IV) Variable Importance Assessment

To construct a model with high predictive performance, this study employs Random Forest and SHAP values (SHapley Additive exPlanations) to assess

the importance of all 15 meteorological variables. This approach allows for a quantitative evaluation of the relative contribution of each variable in predicting three types of extreme weather events—drought, flood, and heat wave—while also accounting for potential nonlinear interactions among variables.

(V) Selection of Important Variables

Based on the results from the importance assessment, key meteorological variables that achieve the highest average rankings across different forecasting tasks are selected. These include, for example, consecutive dry days (cdd), maximum daily precipitation in the month (rx1day), and monthly maximum temperature (tasmax). The selected variables serve as the input feature set for CEEMDAN decomposition and subsequent LSTM modeling.

3.4 Modelling

In order to achieve accurate prediction of various extreme weather events such as drought, flood and heat wave, this study proposed a combined modeling method based on "decomposition-reconstruction-prediction-integration" and constructed the CEEMDAN-SE-LSTM model architecture. This method combines the ability of CEEMDAN to extract multi-scale features of climate data, the efficiency advantage of sample entropy (SE) for signal reconstruction, and the powerful performance of LSTM neural network in modeling nonlinear time series.

In addition, the meteorological data used in this study spans from January 1950 to December 2023, comprising a monthly time series. To ensure temporal consistency in model training and validation, the dataset is partitioned chronologically as follows:

Training and validation set: January 1950 – December 2010 (80%), used for model learning and parameter tuning;

Test set: January 2011 – December 2023 (20%), used to evaluate the model's predictive performance;

Prediction target: future data beyond the available records, starting from 2024.

The whole modeling process includes five main steps:

(I) CEEMDAN decomposes each variable

First, for each selected important meteorological variable (such as cdd, tasmax, rx1day, etc.), the CEEMDAN (Complete Ensemble Empirical Mode Decomposition with Adaptive Noise) method is applied. This technique effectively addresses the common "mode mixing" issue present in traditional EMD decomposition by introducing adaptive white noise at each iteration, resulting in multiple IMF components (Intrinsic Mode Functions) that represent different fluctuation frequencies.

These IMF components can be interpreted as subcomponents of the original meteorological variables across various time scales, including quarterly, annual, and multi-year scales. For instance, certain IMFs exhibit seasonal variations (fluctuating every 3–4 months), while others capture interannual or decadal trends. Through this approach, CEEMDAN enables the decomposition of complex and non-stationary climate signals into more regular and analyzable subsequences, thereby facilitating subsequent modeling tasks.

(II) Sample entropy calculation and IMF reconstruction

Given that each variable typically yields multiple IMF components after decomposition, and considering that some IMFs contribute minimally to the predictive task, sample entropy (SE) is employed to assess the complexity of each IMF. A higher SE value indicates a more irregular sequence, whereas a lower SE suggests a more deterministic or trend-like behavior. Based on the distribution of SE values among the IMFs, components with similar complexity levels are grouped and reconstructed into high-frequency fluctuation sequences and low-frequency periodic sequences, respectively. This process achieves dimensionality reduction, eliminates redundant information, and ultimately enhances modeling efficiency.

(III) LSTM model training and prediction of each reconstructed subsequence

Since CEEMDAN decomposition may yield a large number of IMF components, potentially leading to information redundancy and reduced computational efficiency, this study employs "Sample Entropy" (SE) to assess the complexity of each IMF. Higher SE values indicate more irregular sequences, whereas lower SE values suggest more trend-like behavior.

Based on the distribution of high and low entropy values, IMF components with similar complexity levels are grouped and reconstructed into two types of subsequences: high-frequency components (representing short-term fluctuations) and low-frequency components (reflecting long-term trends). This reconstruction process not only preserves essential features but also effectively reduces the modeling dimensionality.

(IV) Summarizing forecast values and reconstructing the complete meteorological variable sequence

After all subsequences have been predicted, their forecasted values are summed along the time axis to reconstruct the complete forecast sequence for each meteorological variable. For example, the LSTM model generates

predictions for three tasmax subsequences: quarterly fluctuations, annual variations, and multi-year trends. These components are then aggregated over time to form the final tasmax forecast sequence for the upcoming months.

This approach enables the simultaneous capture of abrupt extreme events and long-term background tendencies, thereby enhancing both the stability and accuracy of the forecasting process.

3.5 Model Evaluation

In order to scientifically evaluate the effectiveness of the CEEMDAN-SE-LSTM model proposed in this study for predicting extreme weather events, several classical classification evaluation metrics are employed to comprehensively assess model performance. The advantages of the proposed method are further validated through comparative experiments with baseline models.

This study focuses on binary classification tasks for three types of extreme weather events: drought, flood, and heat wave. Therefore, the following widely used classification evaluation indicators are adopted:

Accuracy: reflects the proportion of total samples that are correctly predicted by the model, indicating its overall predictive capability;

Precision: measures the proportion of predicted "extreme events" that are actually true events;

Recall: indicates the proportion of actual extreme events that are successfully detected by the model;

F1 Score: the harmonic mean of precision and recall, particularly suitable for classification tasks with imbalanced data distributions;

ROC Curve: illustrates the model's classification performance by depicting the relationship between the true positive rate (TPR) and the false positive rate (FPR).

To further validate the effectiveness of the CEEMDAN-SE-LSTM model, three control models are designed for comparison:

Traditional LSTM model: directly uses the original meteorological variable sequences as input without any decomposition or preprocessing;

CEEMDAN-LSTM model: applies CEEMDAN decomposition to meteorological variables and inputs all resulting IMF components into the LSTM for independent modeling, but without sample entropy-based screening;

CEEMDAN-SE-LSTM model (proposed in this study): incorporates a sample entropy (SE) analysis step to select and merge informative IMF components, and only high-information subsequences are fed into the LSTM to enhance modeling efficiency and prediction accuracy.

Through these comparative experiments, the contribution of CEEMDAN decomposition and the SE-based reconstruction mechanism in meteorological data modeling can be systematically evaluated. Additionally, the impact of each processing stage on the final forecasting performance is clarified. The results are presented using various visual formats, including performance comparison tables, ROC curves, and training process plots, to ensure clarity, interpretability, and effective communication of findings.

5. Reference list

[1] Intergovernmental Panel on Climate Change (IPCC). (2023). *Climate change 2023: Synthesis report – Summary for policymakers*. <https://www.ipcc.ch/report/sixth-assessment-report-synthesis-report/>

- [2] Muhammad, M. K. I., Hamed, M. M., Harun, S., & Mohd, N. (2024). Heatwaves in Peninsular Malaysia: A spatiotemporal analysis. *Scientific Reports*, 14, 4255. <https://doi.org/10.1038/s41598-024-42911-1>
- [3] World Meteorological Organization (WMO). (2022, November 11). *Economic losses from extreme weather rocket in Asia*. <https://public.wmo.int/en/media/press-release/economic-losses-extreme-weather-rocket-asia-wmo>
- [4] World Bank. (2021). *Climate risk country profile: Malaysia*. <https://climateknowledgeportal.worldbank.org/country/malaysia/climate-data-projections>
- [6] Camps-Valls, G., Fernández-Torres, M. Á., Cohrs, K. H., Höhl, A., Castelletti, A., Pacal, A., ... Williams, T. (2025). Artificial intelligence for modeling and understanding extreme weather and climate events. *Nature Communications*, 16, 1919. <https://doi.org/10.1038/s41467-025-56573-8>
- [7] World Meteorological Organization (WMO). (2024, April 23). *Climate change and extreme weather impacts hit Asia hard* [Press release]. <https://public.wmo.int/en/media-centre/press-release/climate-change-and-extreme-weather-impacts-hit-asia-hard>
- [8] Verma, S., Srivastava, K., Tiwari, A., & Verma, S. (2023). Deep learning techniques in extreme weather events: A review. *arXiv preprint arXiv:2308.10995*. <https://arxiv.org/abs/2308.10995>
- [9] Islam, T., Mazumder, T., Roni, M. N. S., & Nur, M. S. (2024). A comparative study of machine learning models for predicting Aman rice yields in Bangladesh. *Heliyon*, 10(23), e40764. <https://doi.org/10.1016/j.heliyon.2024.e40764>
- [10] Kumar, V., Kedam, N., Sharma, K. V., Khedher, K. M., & Alluqmani, A. E. (2023). A comparison of machine learning models for predicting rainfall in urban metropolitan cities. *Sustainability*, 15(18), 13724. <https://doi.org/10.3390/su151813724>
- [11] Wang, T., Tu, X., Singh, V. P., Chen, X., Lin, K., & Zhou, Z. (2023). Drought prediction: Insights from the fusion of LSTM and multi-source factors. *Science of The Total Environment*, 902, 166361. <https://doi.org/10.1016/j.scitotenv.2023.166361>
- [12] Zhang, R., Chen, Z.-Y., Xu, L.-J., & Ou, C.-Q. (2019). Meteorological drought forecasting based on a statistical model with machine learning techniques in Shaanxi province, China. *Science of The Total Environment*, 665, 338–352. <https://doi.org/10.1016/j.scitotenv.2019.01.431>
- [13] Li, J., Wu, G., Zhang, Y., & Shi, W. (2024). Optimizing flood predictions by integrating LSTM and physical-based models with mixed historical and simulated data. *Heliyon*, 10(13), e33669. <https://doi.org/10.1016/j.heliyon.2024.e33669>

- [14] Shafiq, F., Zafar, A., Khan, M. U. G., Iqbal, S., Albeshier, A. S., & Asghar, M. N. (2025). Extreme heat prediction through deep learning and explainable AI. *PLOS ONE*, 20(3), e0316367. <https://doi.org/10.1371/journal.pone.0316367>
- [15] Essa, Y., Hunt, H. G. P., Gijben, M., & Ajoodha, R. (2022). Deep learning prediction of thunderstorm severity using remote sensing weather data. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 15, 4004–4013. <https://doi.org/10.1109/JSTARS.2022.3172785>
- [16] Dikshit, A., Pradhan, B., & Huete, A. (2021). An improved SPEI drought forecasting approach using the long short-term memory neural network. *Journal of Environmental Management*, 283, 111979. <https://doi.org/10.1016/j.jenvman.2021.111979>
- [17] Xie, T., Liu, C., Hu, C., Li, W., & Li, R. (2024). Study on long short-term memory based on vector direction of flood process for flood forecasting. *Scientific Reports*, 14, 21446. <https://doi.org/10.1038/s41598-024-72205-5>
- [18] Frifra, A., Maanan, M., Maanan, M., & Rhinane, H. (2024). Harnessing LSTM and XGBoost algorithms for storm prediction. *Scientific Reports*, 14, 11381. <https://doi.org/10.1038/s41598-024-62182-0>
- [19] Huang, N. E., Shen, Z., Long, S. R., Wu, M. C., Shih, H. H., Zheng, Q., ... & Liu, H. H. (1998). The empirical mode decomposition and the Hilbert spectrum for nonlinear and non-stationary time series analysis. *Proceedings of the Royal Society of London. Series A: Mathematical, Physical and Engineering Sciences*, 454(1971), 903–9095. <https://doi.org/10.1098/rspa.1998.0193>
- [20] Torres, M. E., Colominas, M. A., Schlotthauer, G., & Flandrin, P. (2011, May). A complete ensemble empirical mode decomposition with adaptive noise. In *2011 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 4144–4147). IEEE. <https://doi.org/10.1109/ICASSP.2011.5947265>
- [21] Wu, Z., & Huang, N. E. (2009). Ensemble empirical mode decomposition: A noise-assisted data analysis method. *Advances in Adaptive Data Analysis*, 1(01), 1–41. <https://doi.org/10.1142/S1793536909000047>